Users's Manual

West Virginia University
Group Error 404

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Basic Commands

alarm <hour> <minute> <second> <message>

- Creates an alarm at the given time and with the given message
- Time used in hh:mm:ss format (in UTC time)
- Example:

```
mpx> alarm 20 30 0 bedtime
Alarm created.
```

date get

- Displays the current system date
- Example:

```
mpx> date get
02/24/26
```

date set <month> <day> <year>

- Sets the system date in mm/dd/yy format
- Year must be two digits only
- Example:

```
mpx> date set 7 4 20
Date set to: 07/04/20
```

help

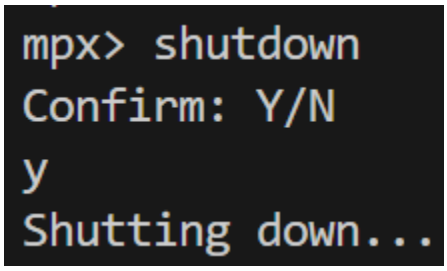
- Shows a list of available commands

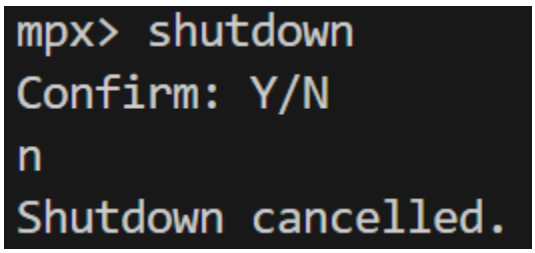
```
mpx> help
Commands:
  date get - Displays the current system date
  date set <month> <day> <year> - Sets the system date mm dd yy
  help - Shows this list of available commands
  shutdown - Shuts down the system and prompts for the confirmation
  time get - Displays the current system time [in UTC time]
  time set <hour> <minute> <second> - Sets the system time hh mm ss [in UTC time]
  version - Shows build/version information
```

shutdown

- Shuts down the system and prompts for the confirmation
- Prompts “Y or N”
- Y will shutdown the system

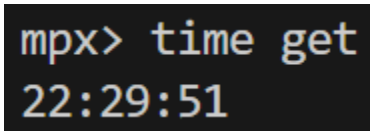
- Anything else will abort shutdown
- Examples:





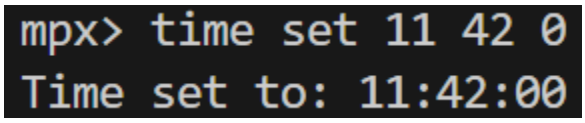
time get

- Displays the current system time (in UTC time)
- Example:



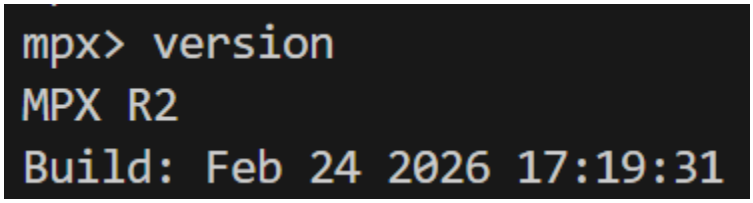
time set <hours> <minutes> <seconds>

- Sets the system time in hh:mm:ss format
- UTC time is used
- Example:



version

- Shows the compilation build time and version
- Example:



PCB Commands

pcb delete <name>

- Deletes an existing process
- The process cannot be a SYSTEM process
- Name is 1-8 characters long
- Example:

```
mpx> pcb delete testpcb1
```

pcb block <name>

- Blocks an unblocked process
- Name is 1-8 characters long
- The process cannot be a SYSTEM process
- Example:

```
mpx> pcb block testpcb1
PCB blocked.
```

pcb unblock <name>

- Unblocks a blocked process
- Name is 1-8 characters long
- Example:

```
mpx> pcb unblock testpcb1
PCB unblocked.
```

pcb suspend <name>

- Suspends an unsuspended process
- Name is 1-8 characters long
- The process cannot be a SYSTEM process
- Example:

```
mpx> pcb suspend testpcb1
PCB suspended.
```

pcb resume <name>

- Resumes a suspended process
- Name is 1-8 characters long
- Example:

```
mpx> pcb resume testpcb1
PCB resumed.
```

pcb set priority <name> <priority>

- Changes the priority level of the process
- Name is 1-8 characters long
- Priority is 0 (high) to 9 (low)
- The process cannot be a SYSTEM process
- Example:

```
mpx> pcb set priority testpcb1 6
PCB priority updated.
```

pcb show name <name>

- Shows all process information
- Name is 1-8 characters long
- Example:

```
mpx> pcb show testpcb1
PCB name: testpcb1
Class: USER
State: READY
Status: NOT_SUSPENDED
Priority: 6
```

pcb show ready

- Shows all processes in the ready queue
- Example:

```
mpx> pcb show ready
READY QUEUE
PCB name: testpcb2
Class: SYSTEM
State: READY
Status: NOT_SUSPENDED
Priority: 2
```

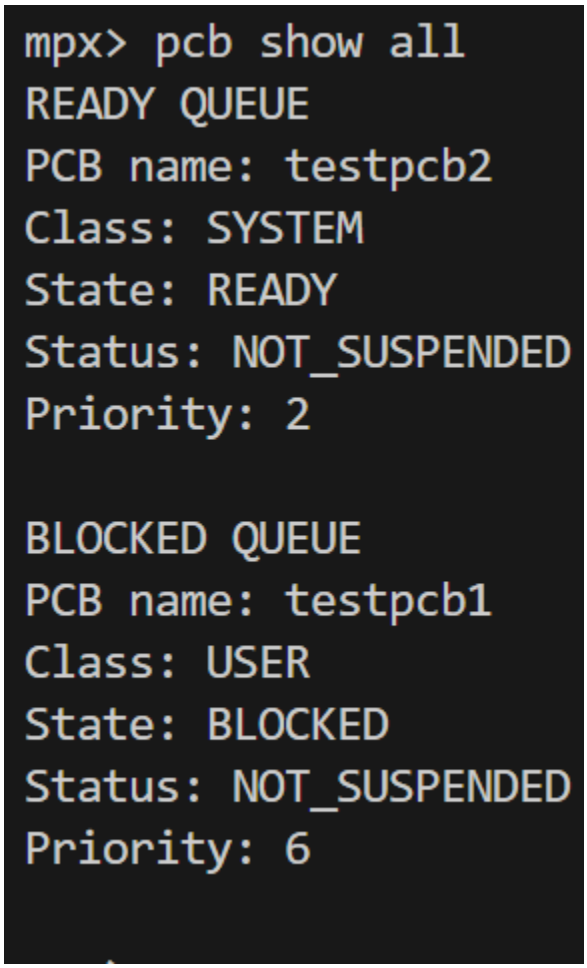
pcb show blocked

- Shows all processes in the blocked queue
- Example:

```
mpx> pcb show blocked
BLOCKED QUEUE
PCB name: testpcb1
Class: USER
State: BLOCKED
Status: NOT_SUSPENDED
Priority: 6
```

pcb show all

- Shows all processes in both queues (ready and blocked)
- Example:



R3 Commands

loadprocess <proc1|proc2|proc3|proc4|proc5>

- Loads one R3 process with names ranging from proc1 to proc5 as ready non-suspended
- Example:

```
mpx> loadprocess proc1
```

loadr3

- Loads all R3 processes as ready non-suspended
- Example:

```
mpx> loadr3
```

loadr3suspended

- Loads all R3 processes as ready suspended
- Example:

```
mpx> loadr3suspended
```

resume all

- Resumes all R3 processes that are ready suspended
- Example:

```
mpx> resume all
```

Memory Commands

allocate memory <decimal size>

- Allocates memory from the heap with the given size in decimal
- The amount of bytes to allocated
- Example:

```
mpx> allocate memory 1024
0xD000010
```

free memory <hex address>

- Frees memory from the heap with the given address of the start of usable memory
- Address in hexadecimal without the 0x prefix
- Example:

```
mpx> free memory 0D000010
```

mem show allocated

- Shows all memory control blocks of allocated memory
- Example:

```
mpx> mem show allocated
MCB allocated
Address: 0D000010
Size: 1024
```

mem show free

- Shows all memory control blocks of allocated memory
- Example:

```
mpx> mem show free
MCB free
Address: 0D000420
Size: 48960
```